

CSS5280 Design and Analysis of Complex Systems

Lecture 9: Classical Agent-based Models:

- **Dynamic Models of Segregation**
- **Competition between Collective and Individual Dynamics**

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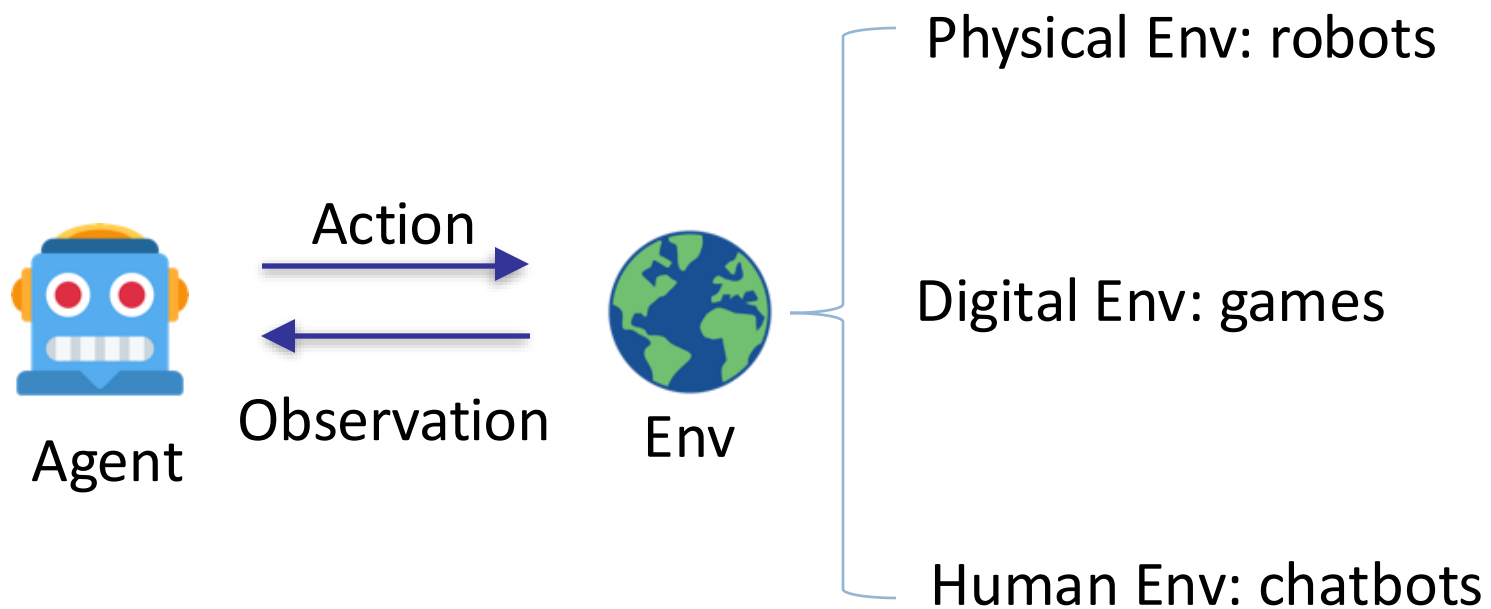
School of Data Science (by courtesy)

CUHK, Shenzhen

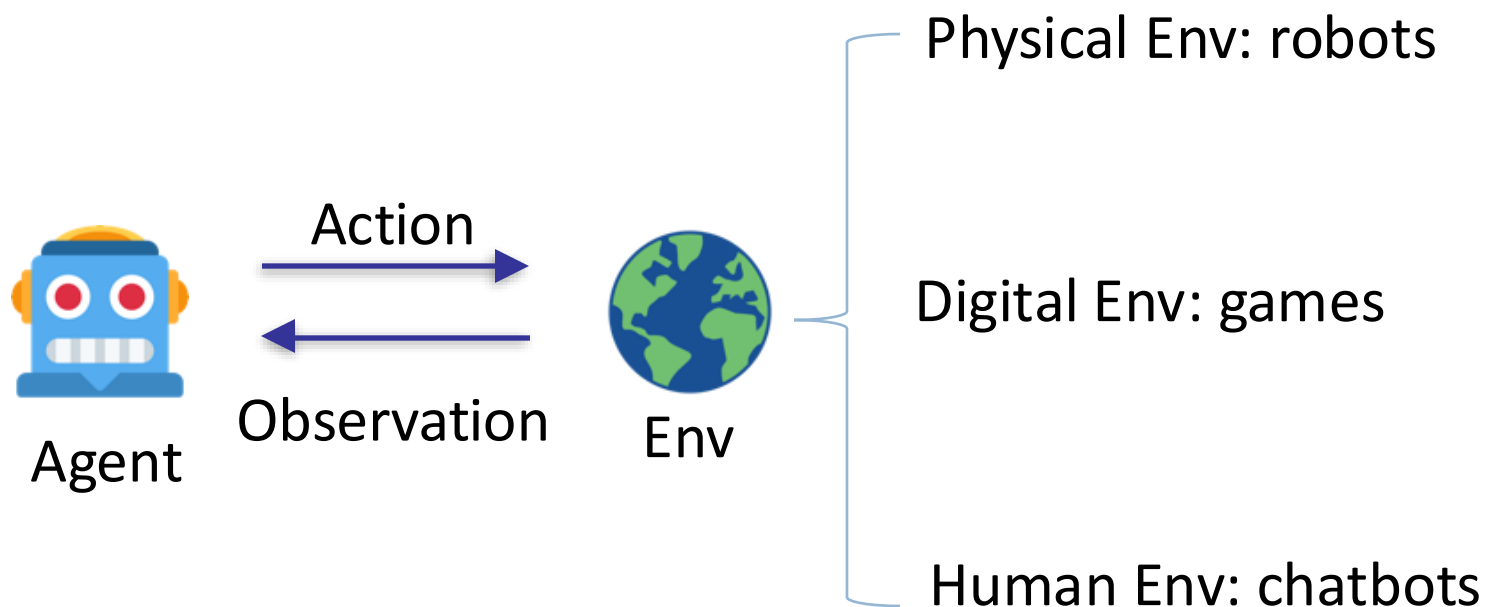
Part 1: What is Agent-based Modeling (ABM) ?

Schelling Model as an Example

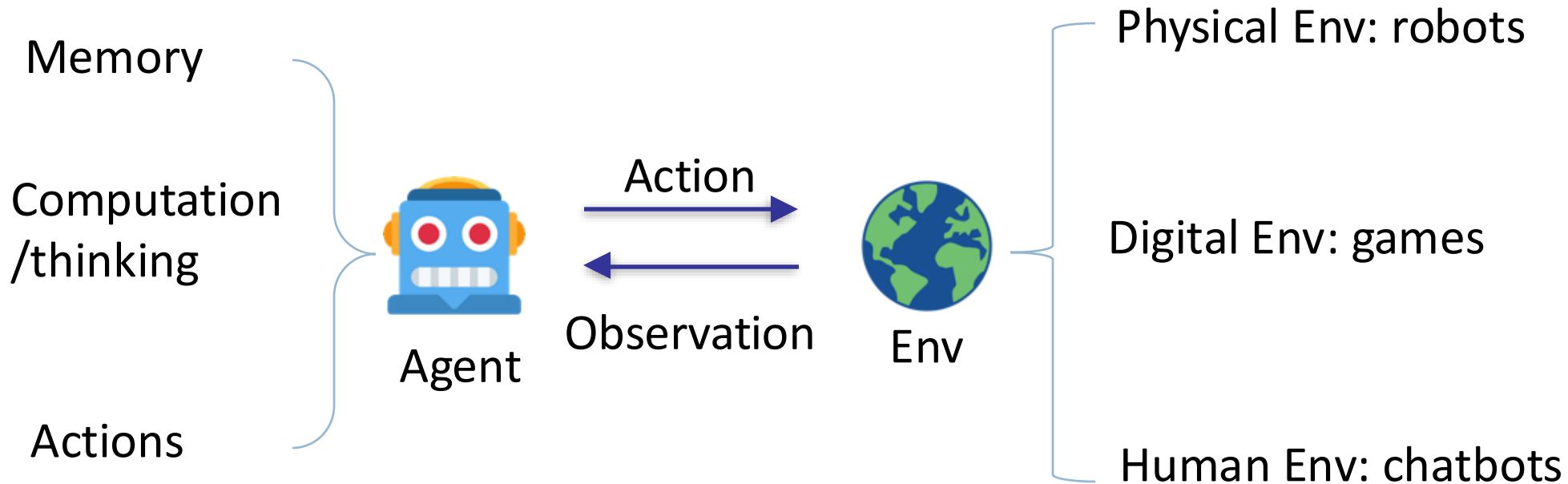
What is “agent”?



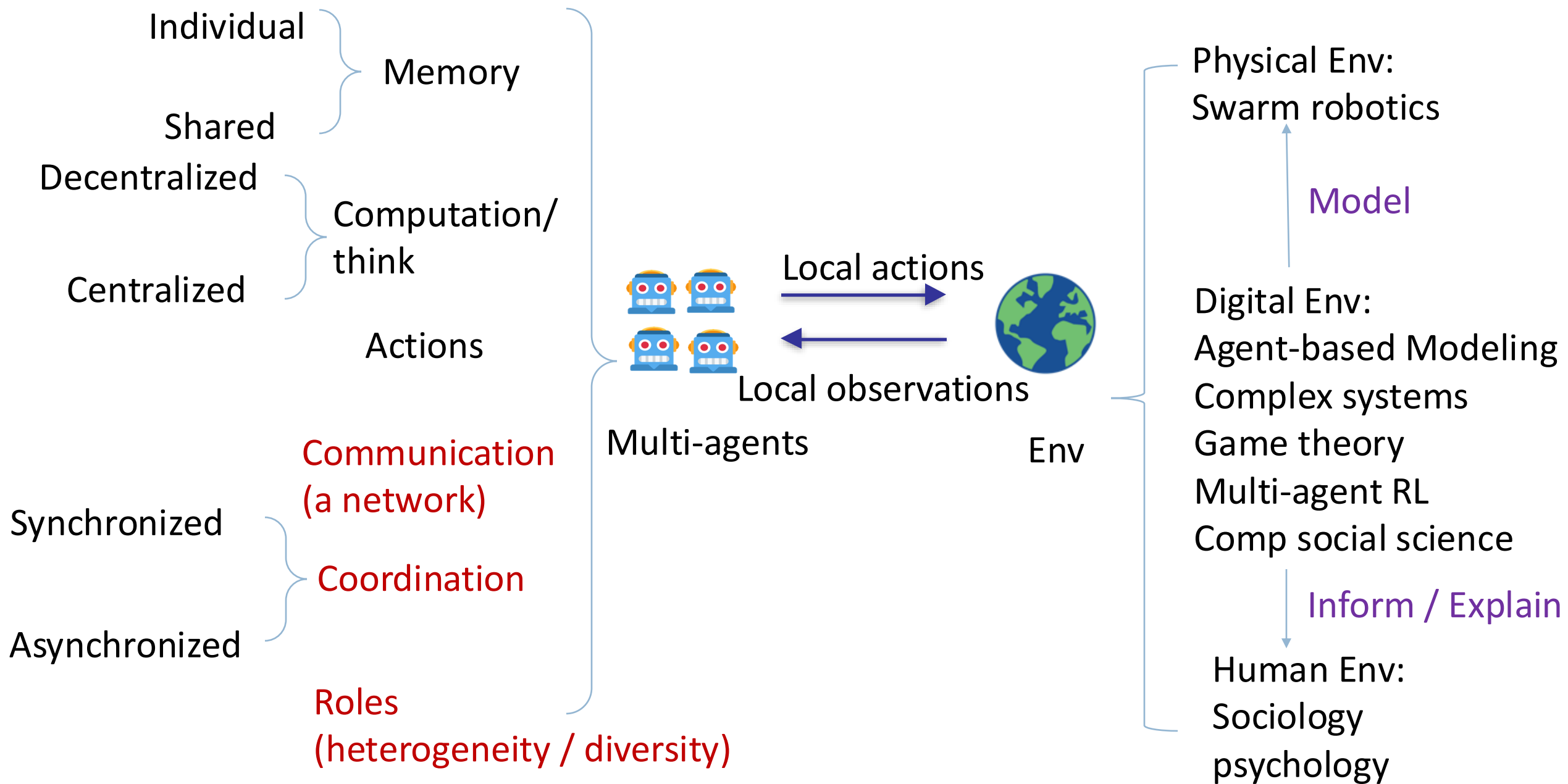
What is “agent”?



What is “agent”?



What is a “multi-agent system”?



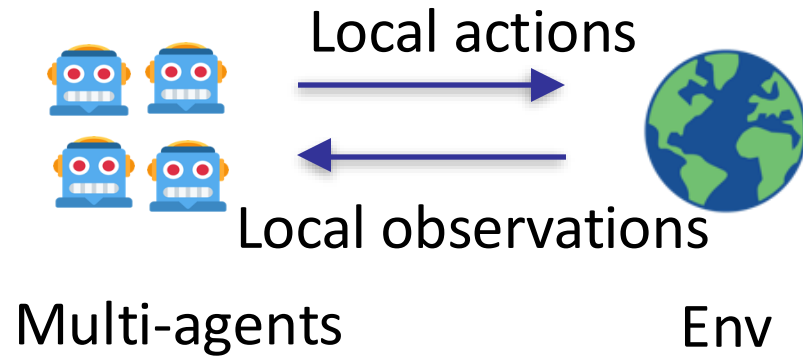
Multi-agent System Coordination

Synchronized vs. Asynchronous Execution:

- **Synchronized:** Agents take turns or follow a predefined execution schedule.
- **Asynchronous:** Agents act independently, without waiting for others, leading to more realistic and dynamic simulations.

Eg. To mimic this uniformly random activation sequence in a local way, we assume each particle has its own Poisson clock with mean 1 and activates after a delay t drawn with probability e^{-t} . After completing its activation, a new delay is drawn to its next activation, and so on.

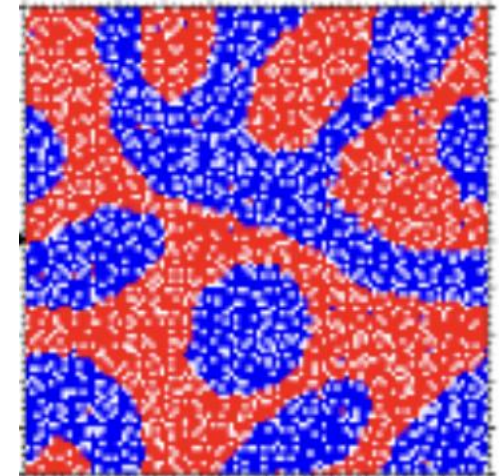
What is a “multi-agent system”?



Physical Env:
Swarm robotics



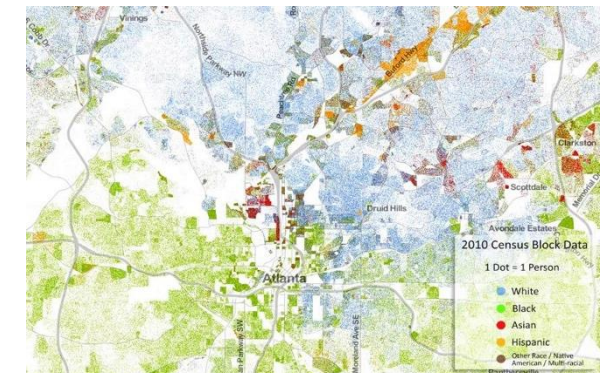
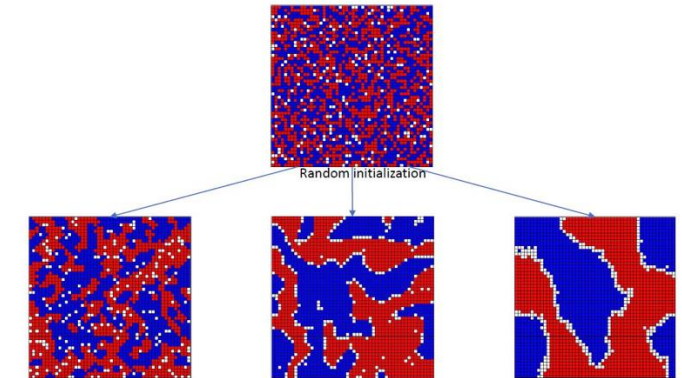
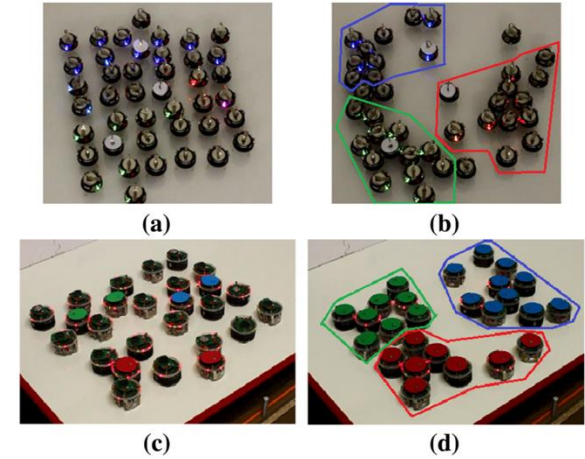
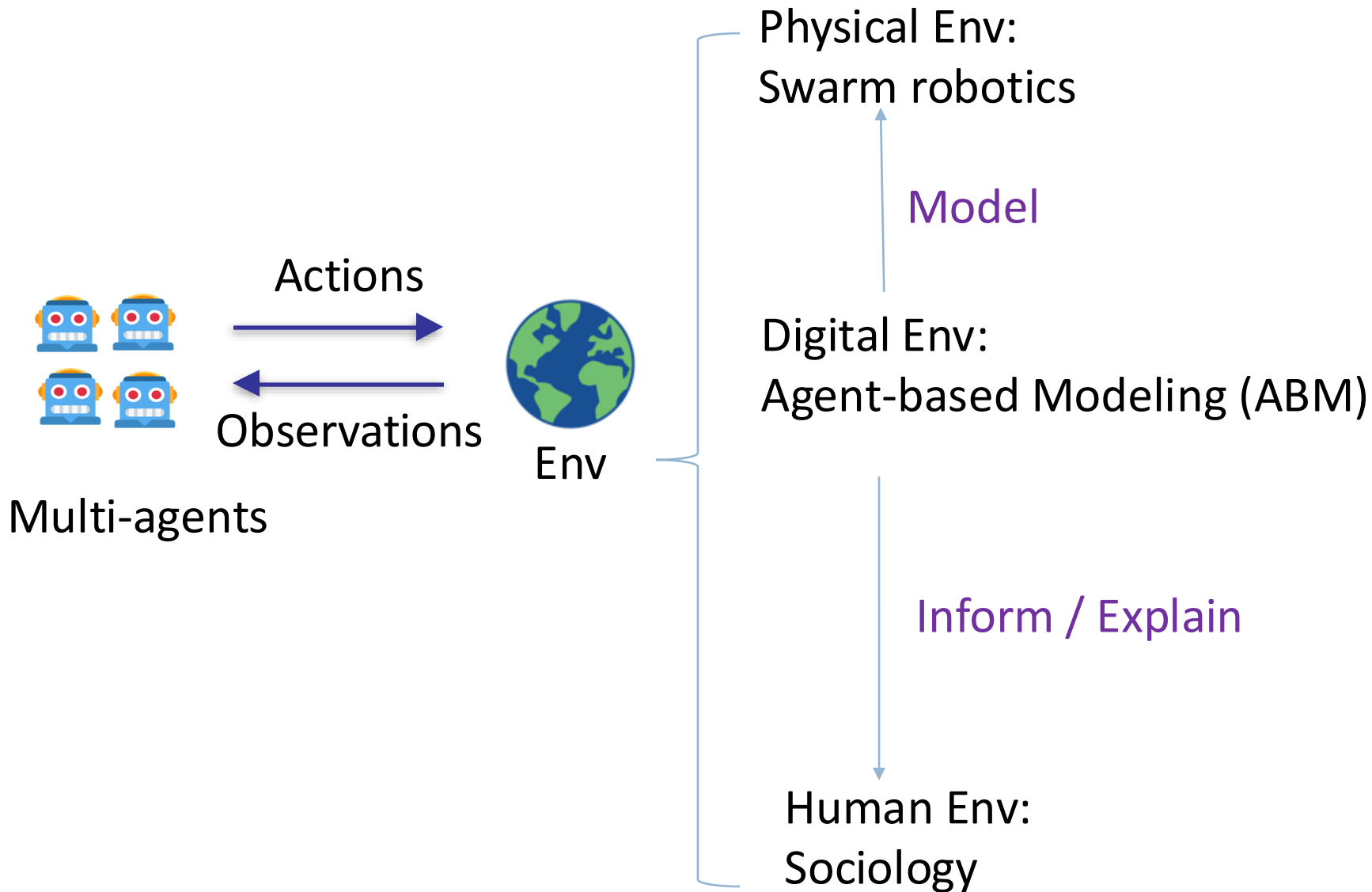
Digital Env:
Agent-based Modeling (ABM)



Human Env:
Sociology

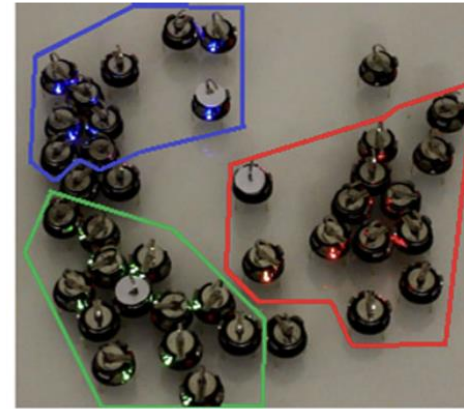


What is a “multi-agent system”?





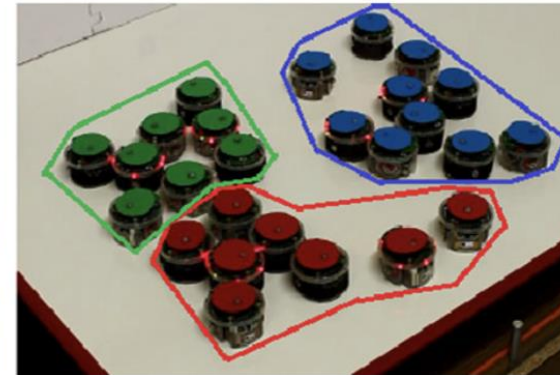
(a)



(b)



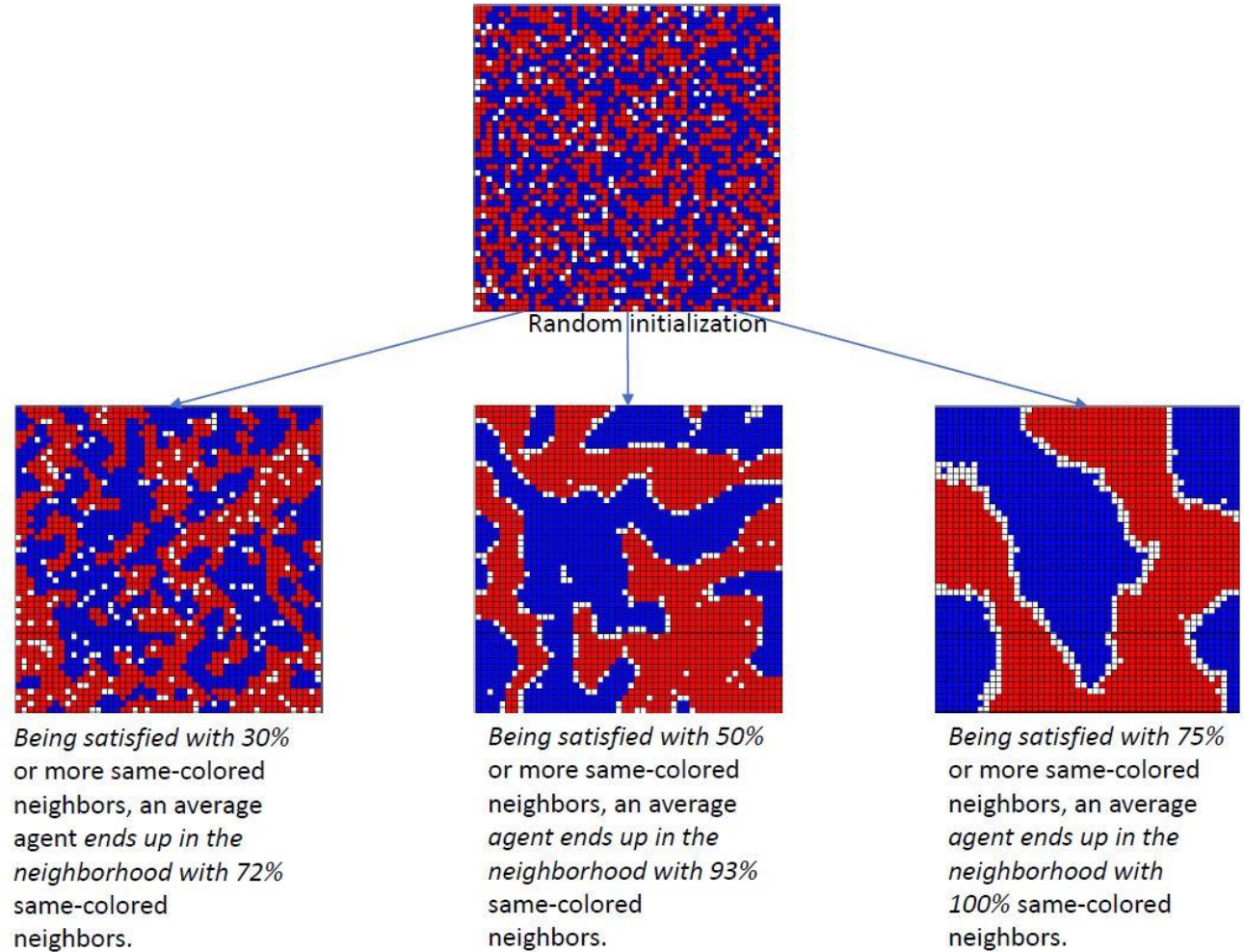
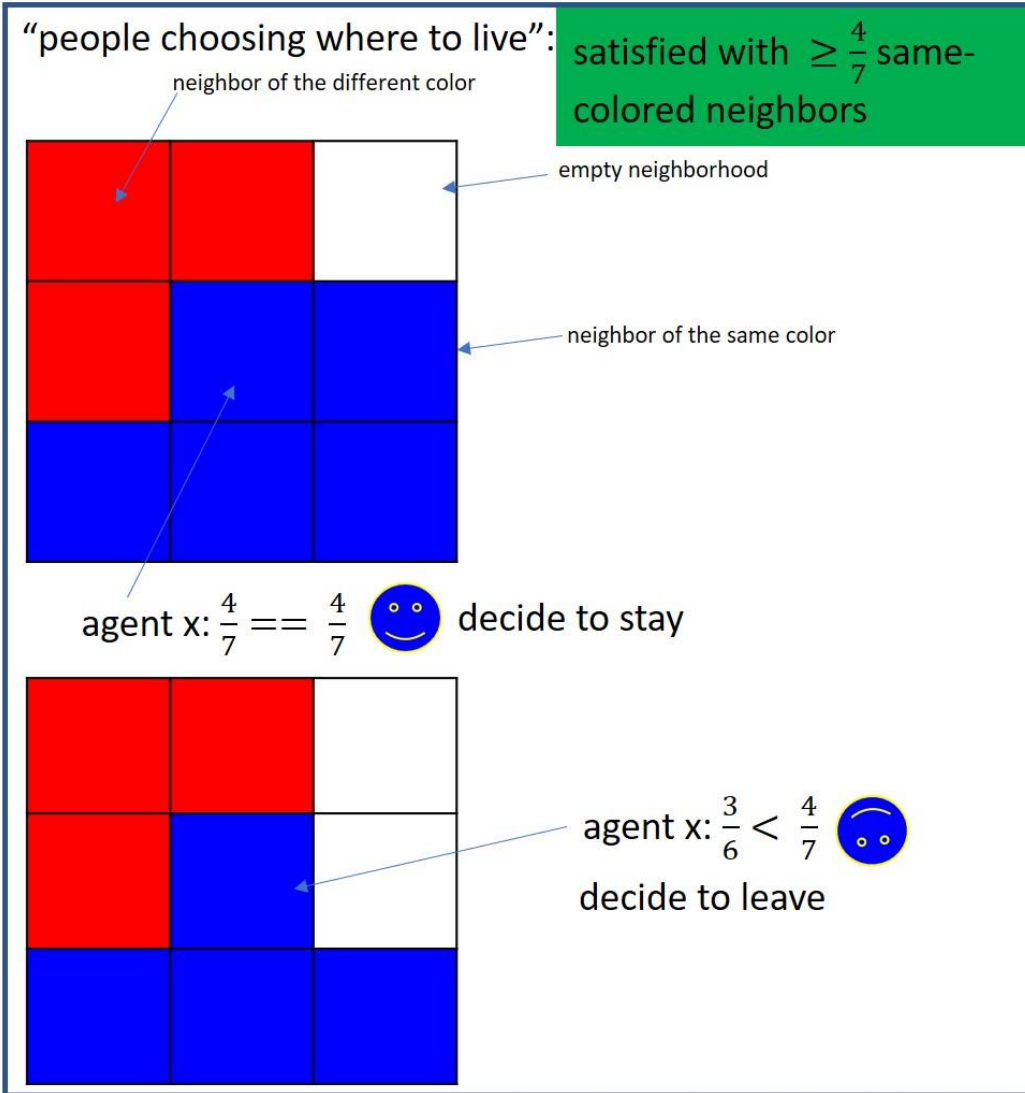
(c)



(d)

Fig. 15 Snapshots from a segregation trial with Kilobots: **a** initial grid formation with three leaders, marked with white tags; **b** result after segregation occurred. Trial with e-pucks: **c** initial grid formation with three pairs of leaders marked with tags; **d** result after segregation occurred, tags were added after the experiments for visualisation

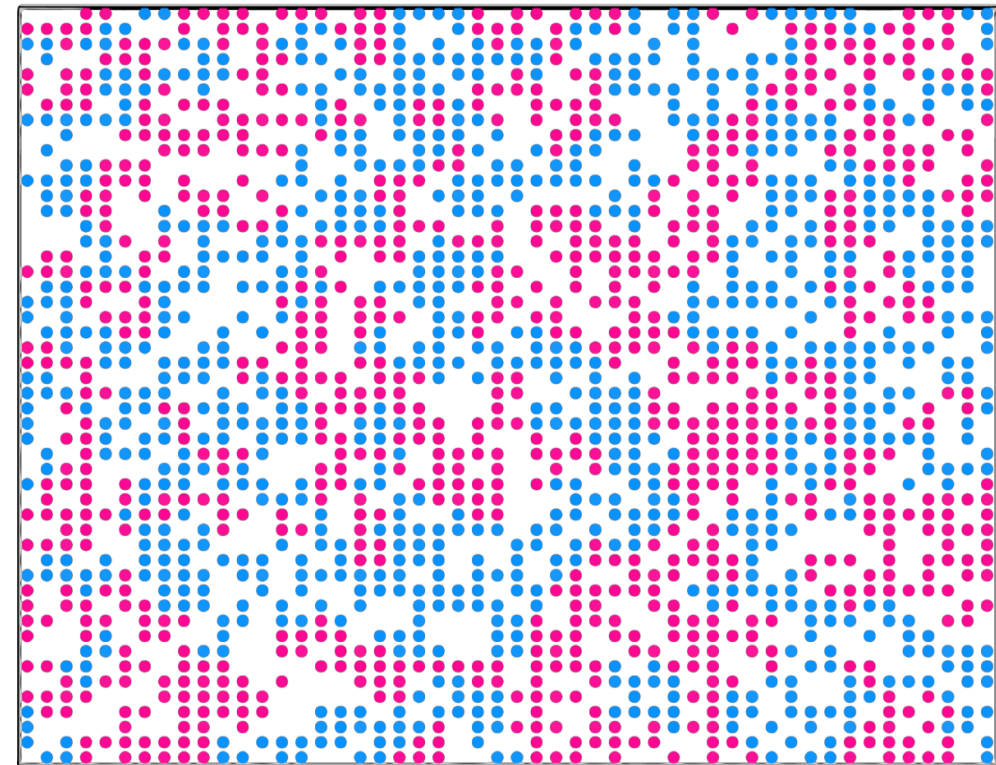
Schelling Model



Example of ABM: Thomas Schelling's Model

- **Rule:** Each agent has a “tolerance” threshold p ;
If fewer than fraction p of its neighbors are of the same color $\rightarrow$ the agent moves to a random vacant site
- **Key Insight:** Even when p is very small (agents are highly tolerant) $\rightarrow$ large-scale **segregation patterns** still emerge!
- **Takeaway: Micro-level rules** (individual preferences) $\rightarrow$ can produce unexpected **macro-level outcomes** (segregated neighborhoods)

Schelling Model with two colours: Final State with Happiness Threshold 30%

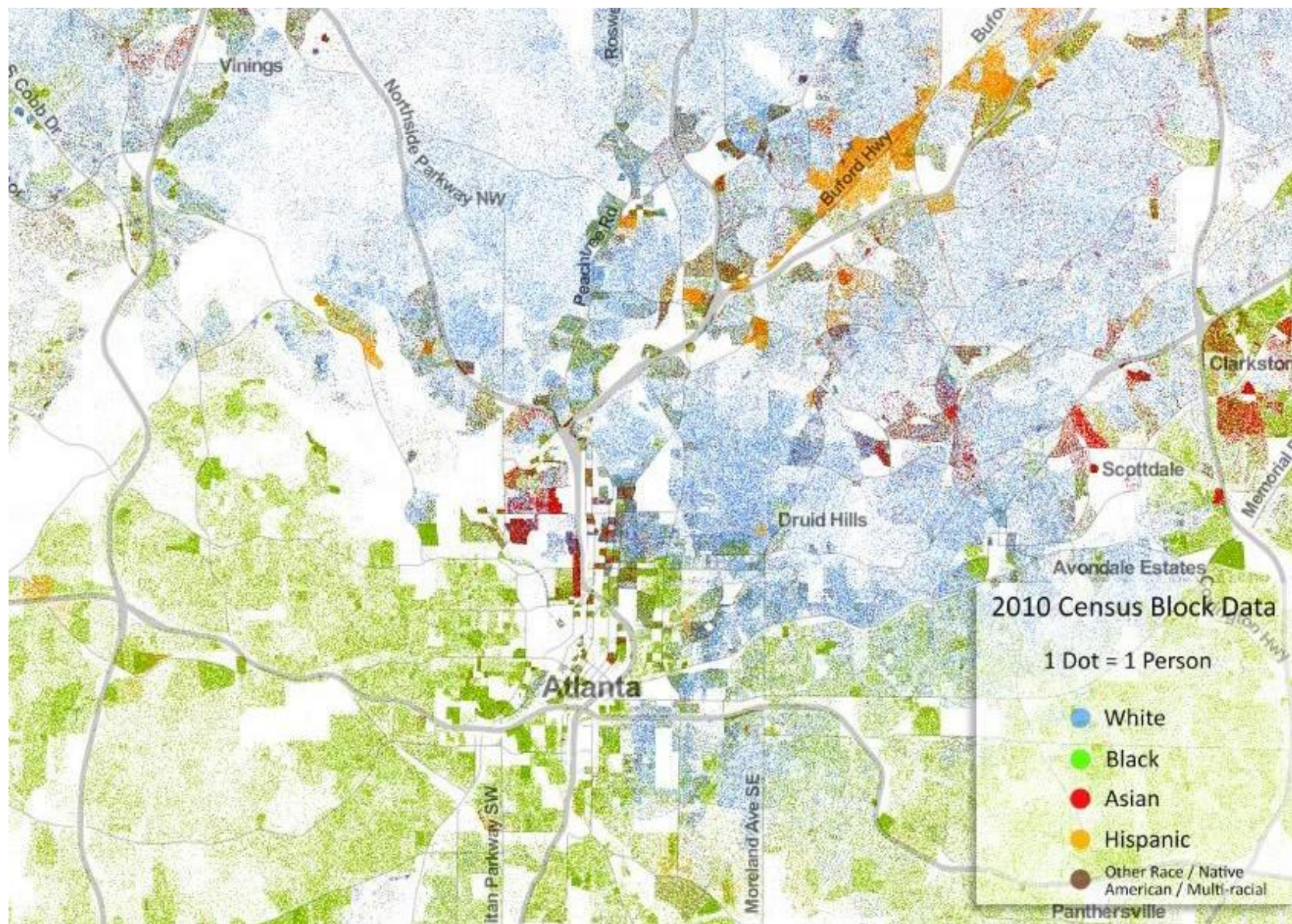


Segregated neighbourhoods

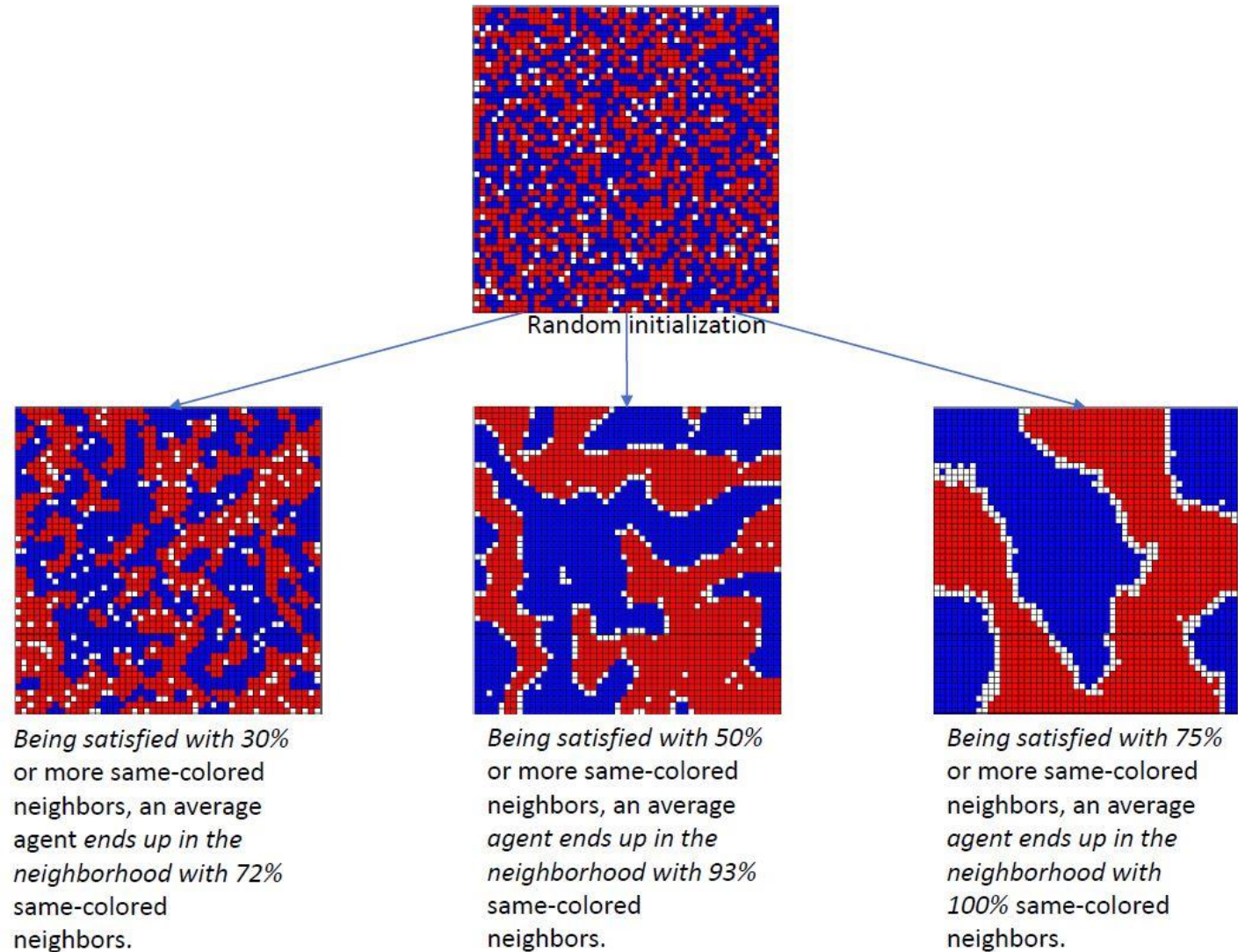
Thomas Schelling (1921–2016)

- American economist and political scientist
- 2005 Nobel Prize in Economics
- Known for:
 - Micromotives and Macrobehavior (1978)**
 - Early models of **segregation dynamics**
 - Illustrating how **simple individual rules** can lead to **complex social patterns**





Emergence refers to a phenomenon where **complex patterns, behaviors, or properties** arise from simple interactions among individual components of a system, without any central control or explicit design. The whole system exhibits properties that **cannot be easily predicted** just by looking at its individual parts.



Key Characteristics of Emergence

- Bottom-Up Formation:
Global patterns emerge from **local interactions** among simple units.
There is **no central controller**—the system **self-organizes**.
- Unpredictability:
The outcome is often **unexpected or more than the sum of its parts**.
The system's behavior **cannot be easily reduced** to the properties of individual components.
- Self-Organization:
The system **adjusts dynamically** without external intervention.
Adaptation and evolution often play a role.
- Multi-Level Effects:
Emergent behavior is usually **observed at a higher level** than the interactions of individual components.

Year	Agent-Based Modeling	Game Theory	Multi-Agent RL (MARL)
1940s	Cellular Automata (Von Neumann & Ulam)	Minimax theorem (Von Neumann, 1928)	-
1950s	-	Nash Equilibrium (Nash, 1950)	-
1970s	Schelling's Segregation Model (1971)	Evolutionary Game Theory (Maynard Smith, 1973)	-
1980s	Boids flocking model (Reynolds, 1986)	Iterated Prisoner's Dilemma (Axelrod, 1984)	Markov Games (Littman, 1994, later in the 90s)
1990s	Sugarscape Model (Epstein & Axtell, 1996)	Expanding economic & social science applications	Independent Q-learning for Multi-Agent Systems (Tan, 1993)
2000s	Large-scale agent simulations (e.g., traffic, economics)	Game Theory applied in AI, auctions, and mechanism design	Cooperative MARL (Claus & Boutilier, 1998), Nash Q-learning (Hu & Wellman, 2003)
2010s	AI-driven ABMs, integration with ML	AI + Game Theory (AlphaGo, 2016)	MADDPG (Multi-Agent Deep Deterministic Policy Gradient, Lowe et al., 2017), Self-Play in AlphaZero (2017)
2020s	Large-scale AI-driven simulations (e.g., social simulations)	Game-Theoretic AI (AlphaZero, OpenAI Five)	MARL for real-world applications (e.g., traffic control, robotics, LLM multi-agent systems)

PARABLE OF THE POLYGONS

A PLAYABLE POST ON THE SHAPE OF SOCIETY

by vi hart + nicky case

[español](#) | [deutsch](#) | [français](#) | [português](#) | [日本語](#) | [中文](#) | [italiano](#)
[magyar](#) | [nederlands](#) | [हिन्दी](#) | [čeština](#) | [Русский](#) | [العربية](#) | [Українська](#) | [עברית](#)



Let's play an online game together: <https://ncase.me/polygons/>

Part 2:

Competition Between Collective and Individual Dynamics

**“Sometimes, everyone chasing their own happiness can
make the whole society worse off.”**

—today, we’ll see exactly why, and when, that happens.

Individual Actions vs. Collective Outcomes

– Why It Matters?

Different academic fields take very different approaches to connecting the **micro** (individual) and **macro** (collective) levels:

- **Physics:** Focuses on *global optimization* – systems move toward the state that minimizes total free energy.
- **Economics:** Focuses on *individual optimization* – each agent seeks to maximize their own utility, leading to a **Nash equilibrium**.

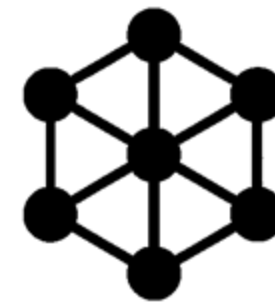
Individual Actions vs. Collective Outcomes

– Why It Matters?

These approaches can produce **radically different results**:

- Global optimization → highest overall “happiness” or efficiency.
- Individual optimization → everyone looks after their own interest, but the collective outcome can be far from ideal.

This paper builds a simple, adjustable model with a cooperation **parameter** α , allowing us to move smoothly between the two perspectives and study the differences in outcomes.



Physics



α



Economics

A Simplified City Model

The “city” is divided into Q blocks, each containing H housing units (cells).

Each housing unit can hold at most **one resident**.

The **density** of block q is: $\rho_q = \frac{n_q}{H}$

where n_q , is the number of residents in block q .

All residents share the **same utility function** $u(\rho_q)$,

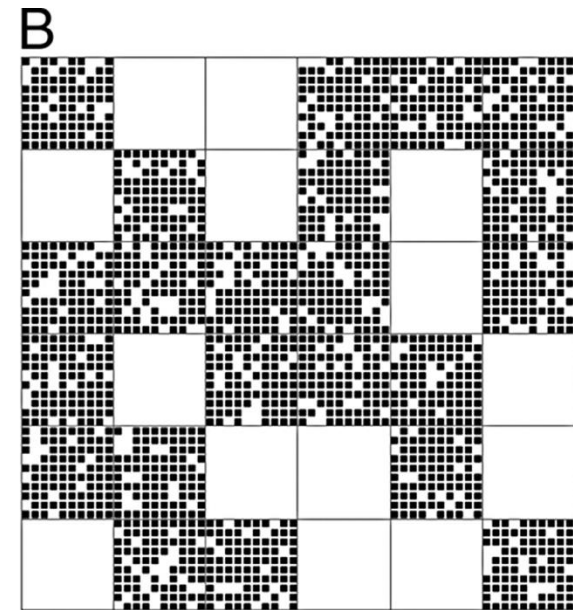
which measures how satisfied they are with the density of their block.

Global utility is the total satisfaction of all residents: $U = H \sum_q \rho_q u(\rho_q)$

This setup lets us connect the *micro-level* (individual preferences) with the *macro-level* (overall city pattern).



Mixed state – residents evenly distributed across blocks.



Segregated state – some blocks overcrowded, others empty.

Move or Stay? Considering Myself and Others

At each step:

1. Randomly pick **one resident** and **one vacant unit**.
2. Calculate the “gain” G from moving.

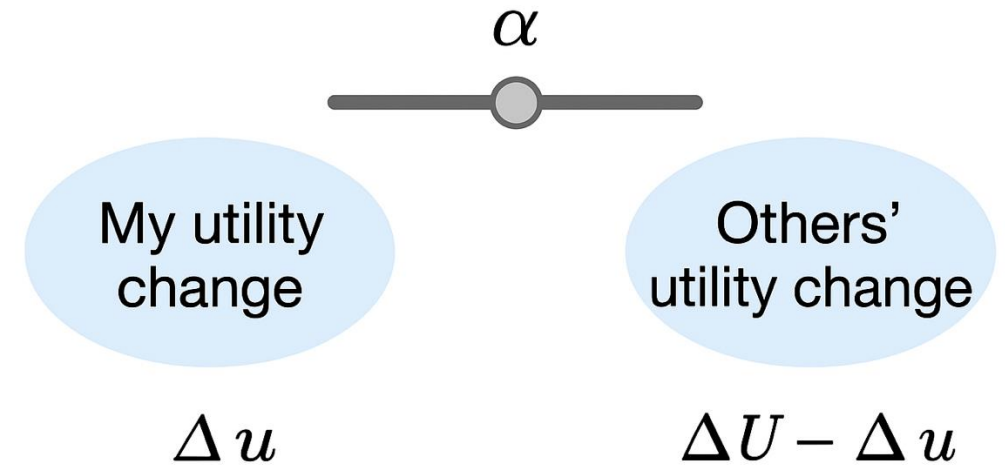
Formula for gain: $G = \Delta u + \alpha(\Delta U - \Delta u)$

- Δu : Change in *my* own utility if I move.
- $\Delta U - \Delta u$: Change in *everyone else's* utility.
- α : Global *cooperation level* shared by all residents.

Special cases:

- $\alpha = 0$: Purely selfish – only care about my own utility.
- $\alpha = 1$: Fully cooperative – only care about total utility.

This setup allows us to smoothly adjust between **economic behavior** (self-interest) and **physical-system-like behavior** (global optimization).



Residents Prefer “Half-Full” Blocks

The model uses an **asymmetrically peaked utility function**:

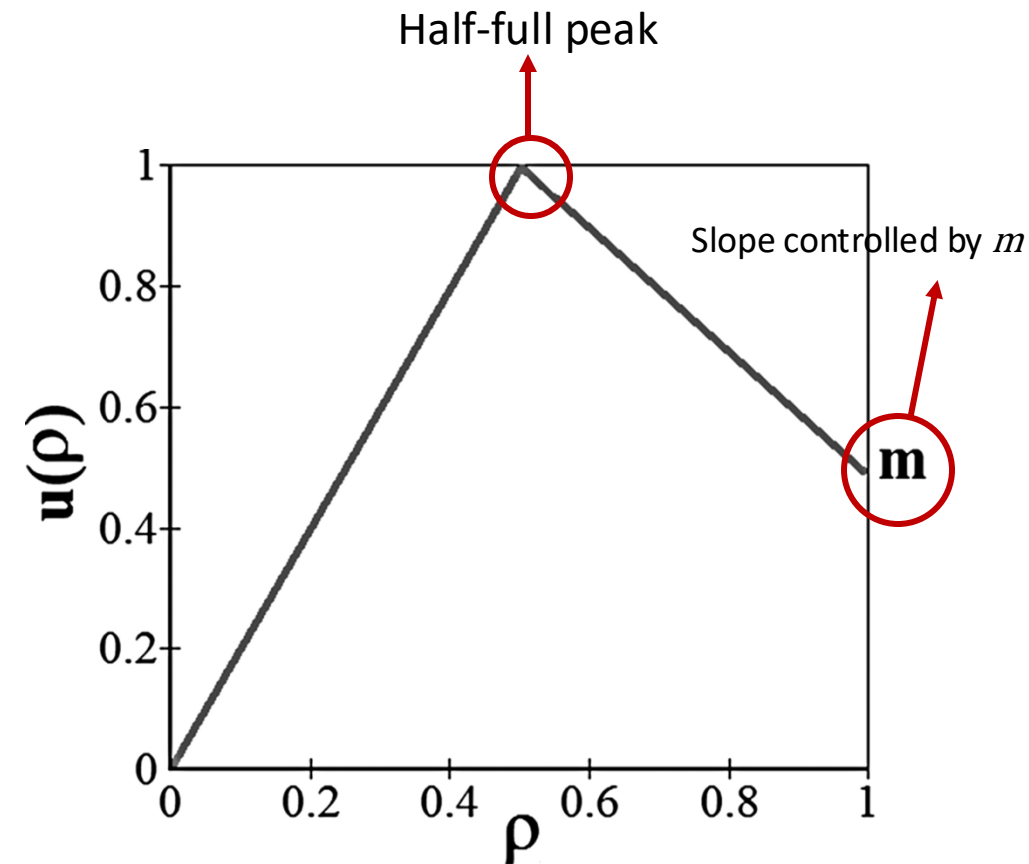
- Utility is **highest** when the block density is $\rho=1/2$ (half-full).
- This reflects a preference for mixed neighborhoods rather than overcrowded or empty ones.

A shape parameter m controls **how fast utility declines** on the crowded side ($\rho > 1/2$):

- Smaller $m \rightarrow$ utility drops quickly as crowding increases.
- Larger $m \rightarrow$ residents tolerate higher density before utility declines.

For the example analysis:

- **Global density** fixed at $\rho_0=1/2$.
- **Temperature** $T \rightarrow 0$ to focus purely on utility-driven decisions without random noise.



From Mixed to Segregated: The Critical Point of α

High cooperation (α large):

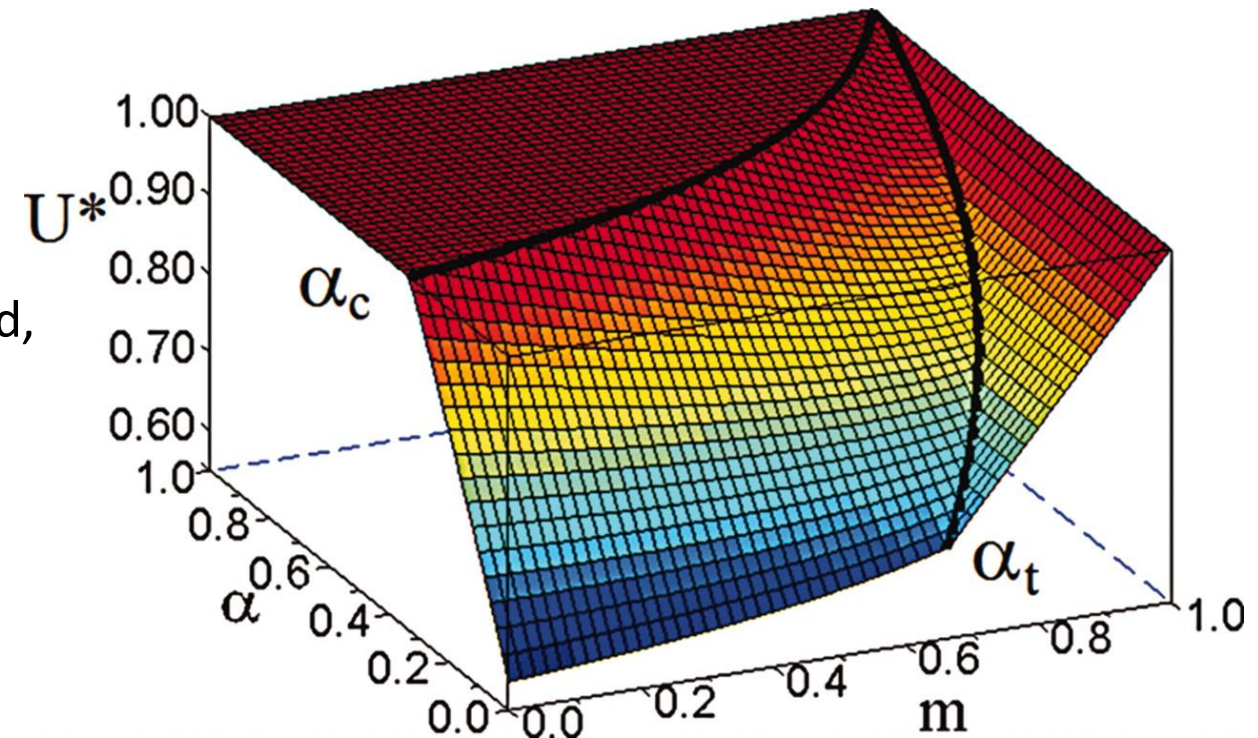
- **Mixed state** – each block has the same density.
- Highest possible global utility.

Low cooperation (α small):

- **Segregated state** – some blocks are overcrowded, others are empty.
- Lower global utility despite individuals trying to improve their own satisfaction.

There is a **critical value**: $\alpha_c = \frac{1}{3 - 2m}$

- If $\alpha > \alpha_c$: mixed state.
- If $\alpha < \alpha_c$: segregated state.



💡 As m increases (more tolerance for crowding), the required α_c for a mixed state becomes higher.

The Role of Noise and Model Robustness

Low temperature (T small):

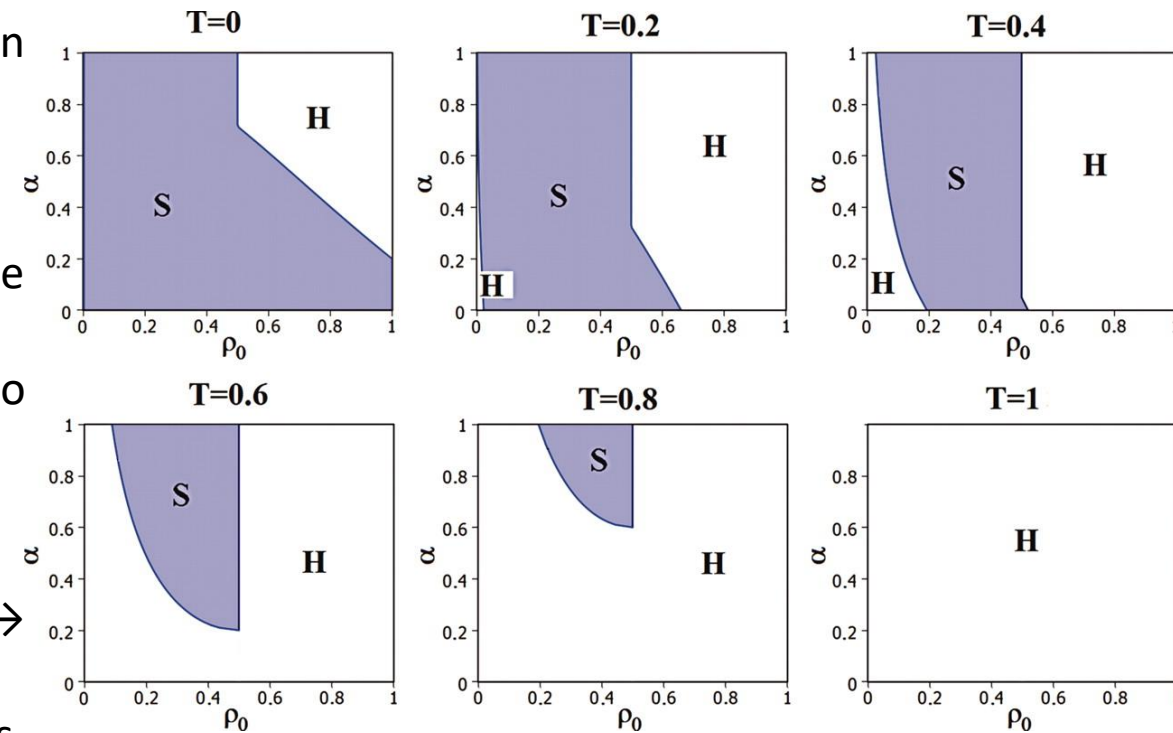
- Noise is weak; the phase diagram and conclusions remain unchanged.

High temperature (T large):

- Randomness dominates $\rightarrow$ city becomes more homogeneous.
- This homogeneity is **not** due to utility optimization but to random movement (“noise-driven mixing”).

Robustness checks:

- Adding two types of residents (e.g., red and green) $\rightarrow$ qualitative results remain the same.
- Changing neighborhood definition (nearest neighbors vs. predefined blocks) $\rightarrow$ qualitative results unchanged.



💡 **Analogy:** Just like stirring coffee ☕ — constant stirring (noise) evens out local differences in concentration.

Economic and Policy Implications

From an **economics perspective**, the cooperation parameter α can be seen as the **strength of policy mechanisms** that internalize externalities (e.g., a *Pigouvian tax*).

Traditional view:

- Only when $\alpha=1$ are all externalities removed, achieving the optimal outcome.

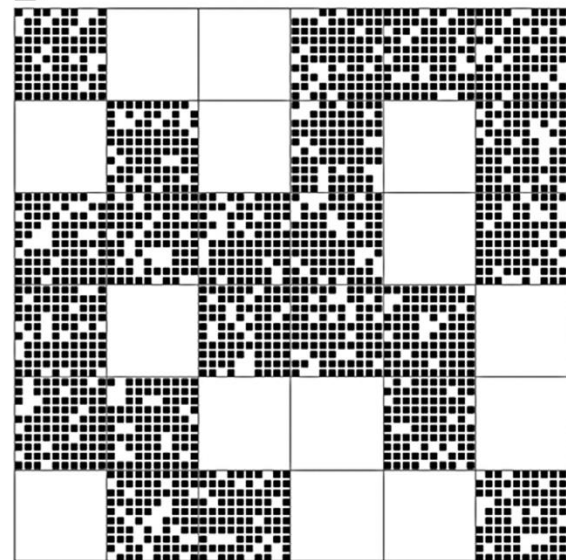
Finding of this paper:

- As long as α exceeds the critical threshold α_c , the system can avoid inefficient segregation and reach an efficient mixed state.
- This means society can achieve **significant improvement** even without completely eliminating all externalities.

A



B



Two Extensions:

1. Catalyst Effect of Altruists

- Even a very small number of altruists can trigger the system into the optimal state.
- “*Small perturbations → Big changes*” in social dynamics.

2. Urban Investment Intervention

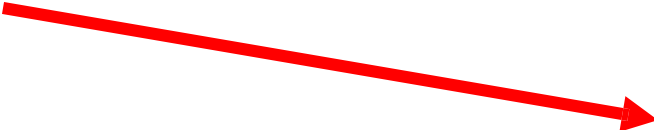
- City planners can reshape utility functions to guide individual choices.
- Shows how **institutional design** can improve social welfare.

Part 3

**If only a small number of people are altruists ,
can they still improve the overall outcome?**

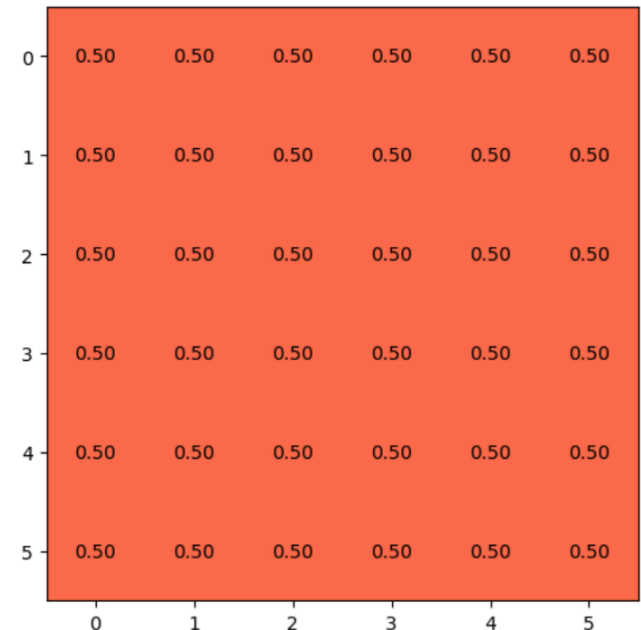
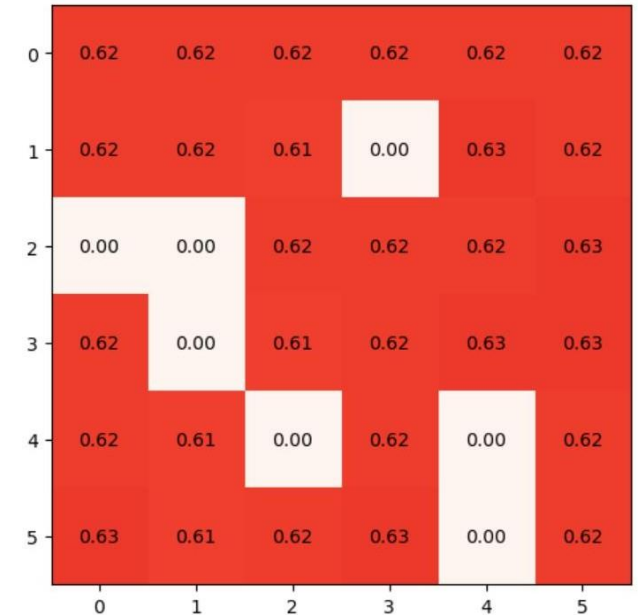
The "Catalyst Effect" of Altruists

From part 2, we learned:

- If everyone is purely self-interested, the city tends to get crowded (low social welfare). 
- If everyone cooperates to some extent ($\alpha > \text{const}$), the city remains mid-level block density (high social welfare). 

But here's the problem:

In reality, not everyone can be cooperative.



Research Question

We want to ask:

- When most people are still self-interested,
- What happens if we mix in a small group of altruists?

Intuition:

- It might help a little, but the effect should be limited.

Actual Finding:

Surprisingly, even a *tiny* fraction of altruists can *catalyze* the entire system into the optimal state! 🎉🎉🎉

The Model

City setup:

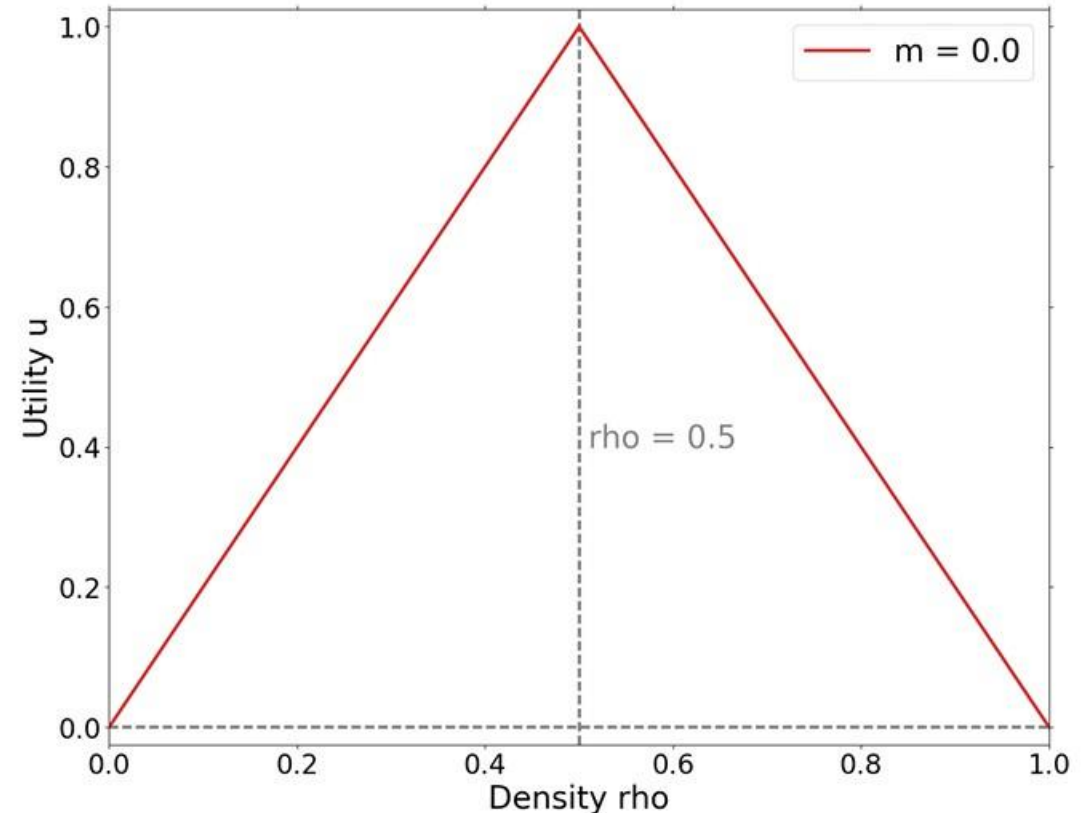
- Q blocks, each with capacity H .

Utility function:

- Utility is highest ($=1$) when density $\rho=0.5$.
- Too sparse ($\rho=0$) or too crowded ($\rho=1$) $\rightarrow$ utility $= 0$.

Two types of agents:

- **Self-interested:** Care only about their own Δu . 🤖
- **Altruists:** Care about the change in *total utility* ΔU . 🧡



Key Result 1: Threshold Effect

If the fraction of altruists in the population is about $p \approx \frac{1}{Q}$,

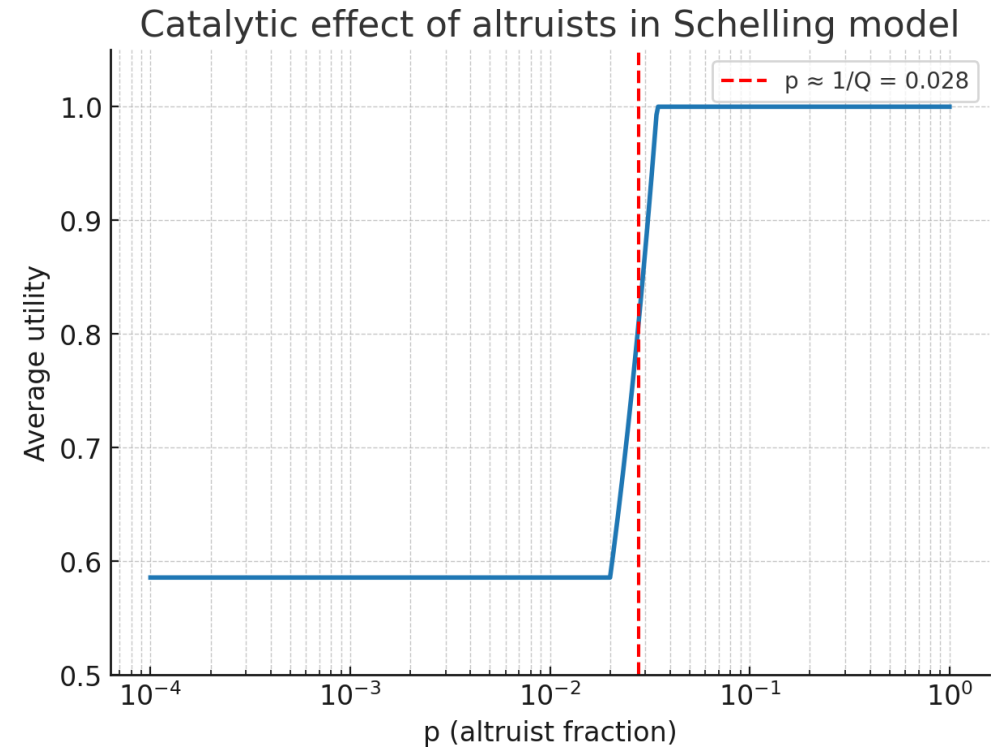
that is already enough to *trigger* the whole system into the optimal mixed state.

Example:

In a city with 36 blocks, only about **3% altruists** are needed.

This is **not** a smooth, linear improvement—

it's a sudden *jump* to high social welfare!



Key Result 2: The Mechanism

Without altruists:

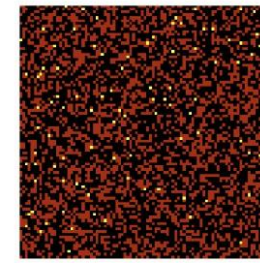
- Self-interested agents cluster together → blocks become overcrowded → low efficiency / social welfare.

With altruists (“sacrifice” behavior):

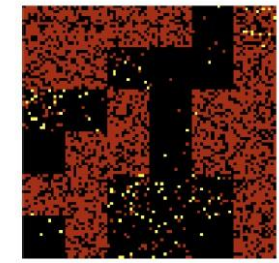
- They *choose* to leave crowded blocks and move to sparser ones.
- This makes sparse blocks gradually more attractive → self-interested agents move in.
- Overcrowded blocks get diluted → overall utility increases.

Bottom line:

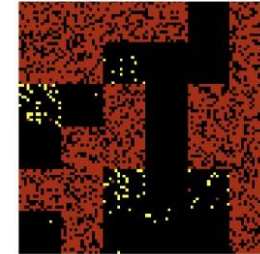
Altruists keep repeating this process, acting like a *catalyst* that drives the whole system toward efficiency / high social welfare.



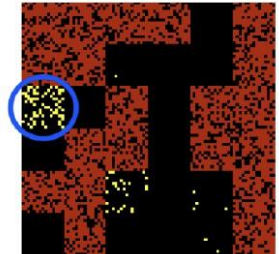
(a) $t = 0$



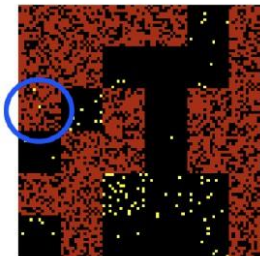
(b) $t = 2$



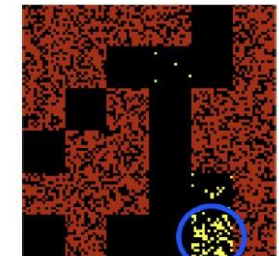
(c) $t = 5$



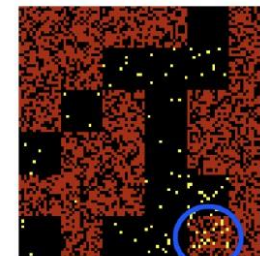
(d) $t = 10$



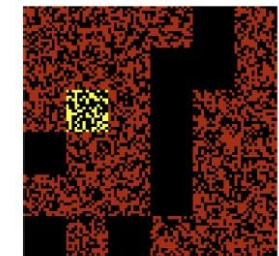
(e) $t = 12$



(f) $t = 35$



(g) $t = 36$



(h) $t = 240$

Why Call It “Catalysis”?

In chemistry:

- A catalyst is *not consumed* but can *accelerate* a reaction.

In the model:

- Altruists don't disappear or “get tired,”
- Yet they continuously guide self-interested agents toward better configurations.

Impact of a single altruist:

- As the system size Q increases,
the *marginal effect* of each altruist becomes even larger.

Summary

Core Finding: A very small number of altruists can *catalyze* the whole system toward the optimal state.

Key Insight:

- Individual self-interest may cause collective failure.
- But just a few “pioneers” — or clever institutional design — can shift the entire system.

Real-World Connections:

- Public policy
- Green behaviors
- Volunteer actions



The altruism of a few may be the *spark* for large-scale social transformation.

Part 4

What if city planners could invest in certain blocks?

Extension: Urban Investment

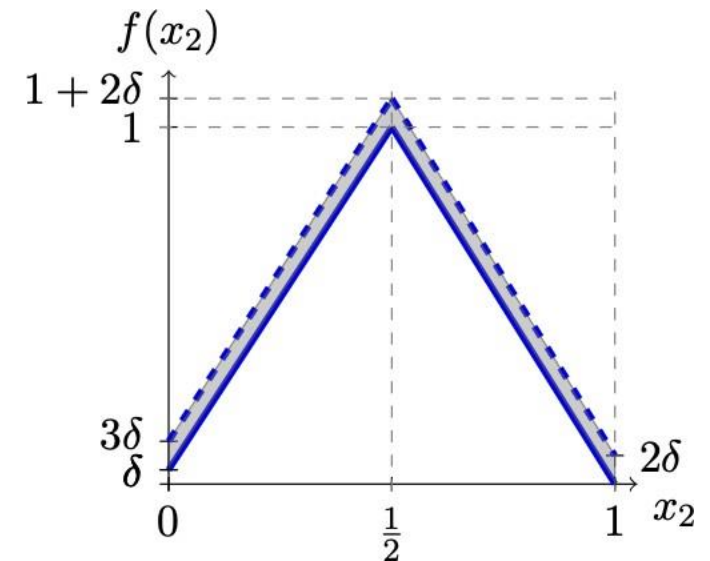
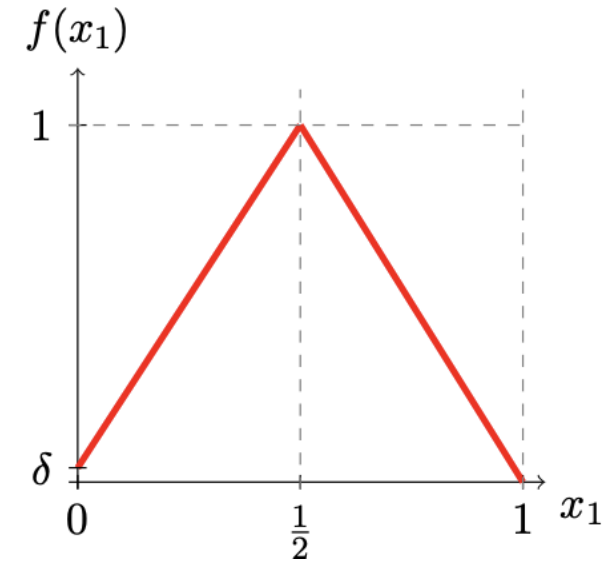
Planner intervention (city-level):

- The planner can change the utility function of certain blocks.
- This makes certain blocks *q* more attractive.

Mathematical expression:

$$g_q(\rho) \geq f_q(\rho)$$

Investment raises the attractiveness of block *q*.



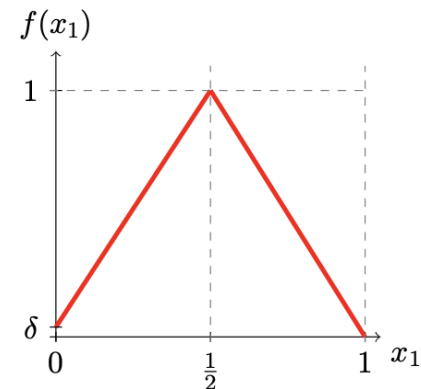
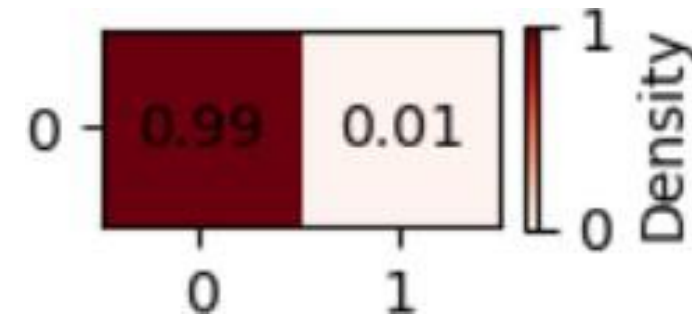
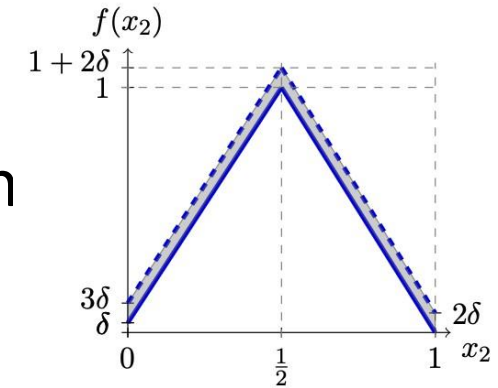
Effects of Investment

Individuals remain self-interested: They only respond to utility differences.

But with investment: Some blocks become more attractive.

Result:

- Migration paths of individuals are indirectly altered.
- At the macro level, social welfare gets changed



Broader Connections: Braess's Paradox

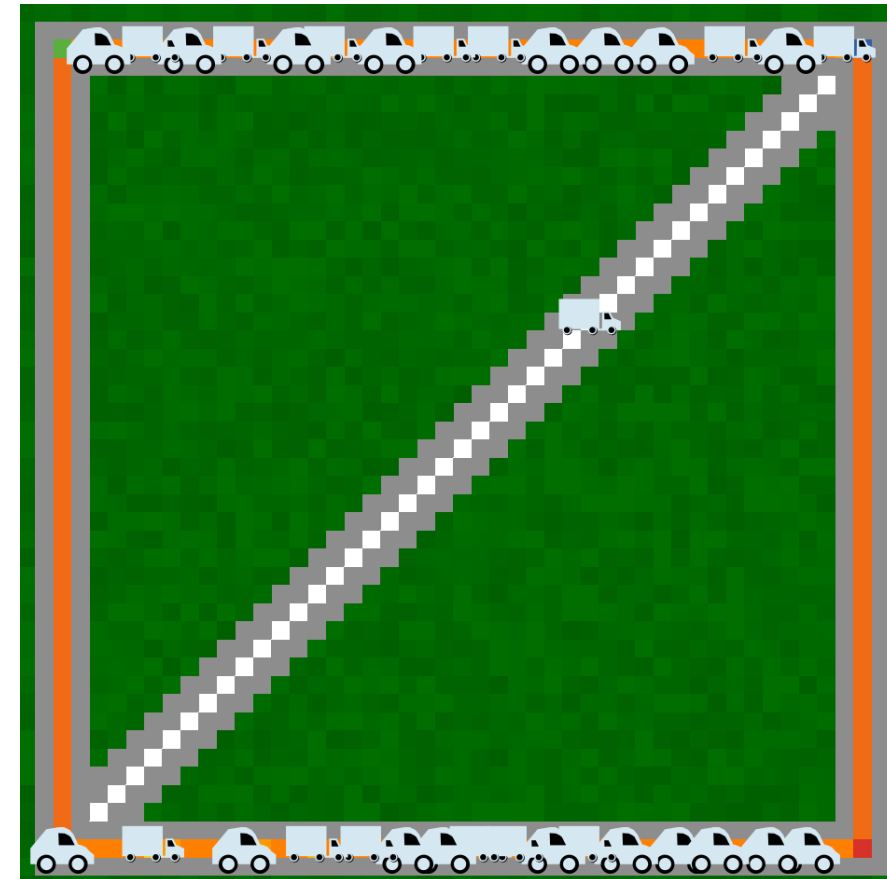
Sometimes, “*better conditions*” can actually make outcomes worse.

Example:

Adding an extra road can make traffic jams *worse*.

Why?

- Each driver chooses what seems best for themselves.
- But collectively, this creates a *less efficient* traffic pattern.



Broader Connections: Congestion Games

Each individual pursues their own *shortest path*.

But when everyone acts this way, the overall outcome is *inefficient (low social welfare)*.

Lesson:

What is *best for individuals* is not always *best for the group*.

Question to think about:

Can we make smart investments to improve social welfare despite egoists (selfish individuals)?

